

Spring 1998

Problem 1 (Sp98). Prove that the polynomial $z^4 + z^3 + 1$ has exactly one zero in the quadrant $\{z = x + iy \mid x, y > 0\}$.

Problem 2 (Sp98). Let f be analytic in an open set containing the closed unit disc. Suppose that $|f(z)| > m$ for $|z| = 1$ and $|f(0)| < m$. Prove that $f(z)$ has at least one zero in the open unit disc $|z| < 1$.

Problem 3 (Fa85, Sp98). Let (M, d) be a nonempty complete metric space. Let S map M into M , and write S^2 for $S \circ S$. Suppose that S^2 is a *strict contraction*; that is, there is a constant $\lambda < 1$ such that for $x, y \in M$, $d(S^2(x), S^2(y)) \leq \lambda d(x, y)$. Show that S has a unique fixed point in M .

Problem 4 (Sp98). Using the properties of the Riemann integral, show that if f is a nonnegative continuous function on $[0, 1]$, and $\int_0^1 f(x) dx = 0$, then $f(x) = 0$ for all $x \in [0, 1]$.

Problem 5 (Sp98). Let A be the ring of real 2×2 matrices of the form $\begin{pmatrix} a & b \\ 0 & c \end{pmatrix}$. What are the 2-sided ideals in A ?

Problem 6 (Sp98). Let G be the group $\mathbb{Q}/\mathbb{Z}$. Show that for every positive integer t , G has a unique cyclic subgroup of order t .

Problem 7 (Sp78, Sp98). Let A and B denote real $n \times n$ symmetric matrices such that $AB = BA$. Prove that A and B have a common eigenvector in $\mathbb{R}^n$.

Problem 8 (Sp98). Let $m \geq 0$ be an integer. Let $a_1, a_2, \dots, a_m$ be integers and let

$$f(x) = \sum_{i=1}^m \frac{a_i x^i}{i!}$$

Show that if $d \geq 0$ is an integer then $f(x)^d / d!$ can be expressed in the form

$$\sum_{i=0}^{md} \frac{b_i x^i}{i!}$$

where the b_i are integers.

Problem 9 (Sp98). Let

$$M_1 = \begin{pmatrix} 3 & 2 \\ 1 & 4 \end{pmatrix} \quad M_2 = \begin{pmatrix} 5 & 7 \\ -3 & -4 \end{pmatrix} \quad M_3 = \begin{pmatrix} 5 & 6.9 \\ -3 & -4 \end{pmatrix}$$

For which i , $1 \leq i \leq 3$, is the sequence M_i^n bounded away from ∞ ? For which i is the sequence bounded away from 0?

Problem 10 (Fa94, Sp98, Sp15). Find the maximum area of all triangles that can be inscribed in an ellipse with semiaxes a and b , and describe the triangles that have maximum area.

Note: See also ??.

Problem 11 (Sp98). Let $A, B, \dots, F$ be real coefficients. Show that the quadratic form

$$Ax^2 + 2Bxy + Cy^2 + 2Dxz + 2Eyz + Fz^2$$

is positive definite if and only if

$$A > 0 \quad \begin{vmatrix} A & B \\ B & C \end{vmatrix} > 0 \quad \begin{vmatrix} A & B & D \\ B & C & E \\ D & E & F \end{vmatrix} > 0$$

Problem 12 (Sp98). Given the fact that $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$, evaluate the integral

$$I = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-(x^2 + (y-x)^2 + y^2)} dx dy$$

Problem 13 (Sp98). Let a be a complex number with $|a| < 1$. Evaluate the integral

$$\int_{|z|=1} \frac{|dz|}{|z-a|^2}$$

Problem 14 (Sp98). Let K be a real constant. Suppose that $x(t)$ is a positive differentiable function satisfying $x'(t) \leq Kx(t)$ for $t \geq 0$. Prove that $x(t) \leq e^{Kt}x(0)$ for $t \geq 0$.

Problem 15 (Sp98). For continuous real valued functions f, g on the interval $[-1, 1]$ define the inner product $\langle f, g \rangle = \int_{-1}^1 f(x)g(x) dx$. Find that polynomial of the form $p(x) = a + bx^2 - x^4$ which is orthogonal on $[-1, 1]$ to all lower order polynomials.

Problem 16 (Sp98). Let $a > 0$. Show that the complex function

$$f(z) = \frac{1 + z + az^2}{1 - z + az^2}$$

satisfies $|f(z)| < 1$ for all z in the open left half-plane $\Re z < 0$.

Problem 17 (Sp98). Let A be an $n \times n$ complex matrix with $\text{tr } A = 0$. Show that A is similar to a matrix with all 0's along the main diagonal.

Problem 18 (Sp98). Let N be a nilpotent complex matrix. Let r be a positive integer. Show that there is a $n \times n$ complex matrix A with

$$A^r = I + N$$